

Squaregles

WHAT'S IN THE KIT

- Squaregles Starter Set
- Drop n Jump Racetrack expansion set



QUICK-START GUIDE

1. Panels connect magnetically — build walls, boxes, and towers.
2. Add the racetrack pieces to route marbles through builds.
3. Challenge: the slowest possible marble journey.
4. Magnets make cleanup genuinely fast.

FULL LESSONS IN THIS GUIDE

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PRO TIP: The 'slowest marble' challenge flips expectations and produces the best engineering talk.

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LESSON · GRADES K-2

Marble Coaster Engineers

LEARNING OUTCOMES

- Students will build a marble path that works from start to finish
- Students will diagnose and fix a failure point
- Students will use cause-and-effect language to explain fixes

MATERIALS

- Squaregles starter set + Drop n Jump expansion
- Marbles (with a strict marble-count routine)
- Builder mats per group

INSTRUCTIONS

1. Demo a 3-piece run; roll the marble and narrate its journey.
2. Groups build their own runs — any design that gets the marble to the bottom.
3. When marbles stick, name it a 'stuck spot' and study it before touching.
4. Fix one thing, retest, repeat until the run completes.
5. Group tour: each team demonstrates and explains their trickiest fix.
6. Magnet-cleanup race to close.

ACTIVITY

Marble Coaster Engineers — build runs, find the stuck spots, fix them one change at a time. Cause-and-effect reasoning with feedback that arrives in two seconds.

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LESSON · GRADES 3-5

Slow Marble Challenge

LEARNING OUTCOMES

- Students will engineer for a counterintuitive goal (slowness)
- Students will time trials accurately and run fair tests
- Students will connect design features to energy and friction concepts

MATERIALS

- Squaregles sets with racetrack expansion
- Marbles and stopwatches
- Trial data table
- Fixed footprint boundary (tape square) for fairness

INSTRUCTIONS

1. Flip the challenge: the SLOWEST marble to reach the bottom wins.
2. Set fair-test rules: same marble, same drop point, run must complete.
3. Teams build, time three runs, and record the average.
4. Iterate: zigzags, gentle slopes, jump landings — what buys the most seconds?
5. Finals: each team's best run timed publicly.
6. Debrief: where did the marble's energy go? Name friction's role.

ACTIVITY

Slow Marble Challenge — winning by losing: the slowest complete run takes the crown. Averaging three timed trials brings fair-testing rigor to a giggling classroom.